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$$\int \frac{xdx}{\sqrt{7-x^2}}$$

$$\text{Stel } t = 7 - x^2 \Rightarrow dt = -2xdx \Rightarrow -\frac{1}{2}dt = xdx$$

$$= -\frac{1}{2} \int \frac{dt}{\sqrt{t}}$$

$$= -\frac{1}{2} \int t^{\frac{1}{2}}$$

$$= -\frac{1}{2} \cdot \frac{2\sqrt{t}}{1} + C$$

$$= -\sqrt{7-x^2} + C$$

References